

Short Paper:

Work Breakdown Structure for the

Redevelopment of the Sanford B. Ladd School

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Developing a Work Breakdown Structure (WBS) is an essential part of defining project scope and organizing the major components of a complex initiative. For the redevelopment of the Sanford B. Ladd School into a community resource hub, the WBS provides a structured hierarchy that supports planning, coordination, and accountability across construction, environmental review, community engagement, and program development. Because the project involves historic preservation, federal grant compliance, and significant building rehabilitation, a clear WBS is necessary to manage parallel workstreams and ensure alignment with regulatory and funding requirements (City of Kansas City, Missouri, 2024; U.S. Department of Housing and Urban Development, 2023; U.S. Economic Development Administration, 2023). The WBS also reflects the technical and architectural considerations documented in the Ladd School redevelopment materials, which guide the scope of renovation and site improvements (Drexel Tech, 2026; LocalCode, n.d.). By decomposing the project into phases, work packages, and activities, the WBS establishes the foundation for effective scope management and supports the development of realistic schedules, resource plans, and performance expectations.

To illustrate the full scope of the redevelopment effort, the following text-based Work Breakdown Structure (WBS) outline presents the project in a hierarchical format that organizes phases, work packages, and activities.

Work Breakdown Structure (WBS) Outline

Project: Redevelopment of the Sanford B. Ladd School into a Community Resource Hub

1.0 Pre-Development Phase

1.1 Community Engagement

- 1.1.1 Conduct neighborhood listening sessions
- 1.1.2 Conduct interest gauge / pre-recruitment drive
- 1.1.3 Facilitate stakeholder interviews
- 1.1.4 Compile community needs assessment

1.2 Governance Formation

- 1.2.1 Establish community steering committee
- 1.2.2 Develop cooperative or land trust framework
- 1.2.3 Approve governance charter

1.3 Funding and Grant Preparation

- 1.3.1 Prepare HUD CDBG application
- 1.3.2 Prepare CCED Sales Tax grant application
- 1.3.3 Prepare EDA Public Works grant application

1.4 Environmental & Architectural Preparation

- 1.4.1 Conduct environmental review and testing
- 1.4.2 Conduct structural analysis and landscape analysis
- 1.4.3 Complete architectural and landscape schematic design
- 1.4.4 Finalize construction documents

2.0 Building Renovation Phase

2.1 Site Preparation & Abatement

- 2.1.1 Perform hazardous materials abatement
- 2.1.2 Complete selective demolition
- 2.1.3 Secure site and install temporary utilities

2.2 Exterior Restoration

- 2.2.1 Repair masonry and façade
- 2.2.2 Replace or restore windows and doors

2.2.3 Repair roofing and drainage systems

2.3 Mechanical, Electrical, and Plumbing (MEP)

2.3.1 Install HVAC systems

2.3.2 Upgrade electrical service and wiring

2.3.3 Replace plumbing and install fixtures

2.4 Interior Construction & Finishes

2.4.1 Build interior walls and program spaces

2.4.2 Install flooring, ceilings, and finishes

2.4.3 Install technology infrastructure and equipment

2.5 Final Inspections & Occupancy

2.5.1 Conduct code compliance inspections

2.5.2 Complete punch list corrections

2.5.3 Obtain certificate of occupancy

3.0 Community Garden Development

3.1 Site Preparation

3.1.1 Clear and grade elevated field

3.1.2 Install/repair fencing and access paths

3.2 Garden Construction

3.2.1 Build raised garden beds

3.2.2 Install irrigation system

3.2.3 Deliver soil and compost

3.3 Community Planting

3.3.1 Organize volunteer planting days

3.3.2 Plant initial crops

3.3.3 Set up maintenance schedule

4.0 Administrative & Program Development (Parallel Workstream)

4.1 Staffing & Volunteer Recruitment

4.1.1 Recruit educators and program staff

4.1.2 Recruit volunteers for pantry and garden

4.1.3 Conduct background checks and onboarding

4.2 Program & Partnership Development

- 4.2.1 Finalize partnership with Parks & Recreation
- 4.2.2 Finalize partnership with Kansas City Public Library
- 4.2.3 Develop curricula for job training and community education
- 4.2.4 Coordinate donation drives

4.3 Operational Readiness

- 4.3.1 Develop policies and procedures
- 4.3.2 Create safety and emergency plans
- 4.3.3 Prepare opening-day operations plan

5.0 Project Closeout

5.1 Grand Opening

- 5.1.1 Coordinate opening event
- 5.1.2 Launch job center and programming
- 5.1.3 Launch pantry and closet
- 5.1.4 Celebrate first garden harvest

The Microsoft Project image that follows provides a detailed software-based representation of the hierarchical WBS outlined in the preceding text. While the written outline presents the project's phases, work packages, and activities at a structural level, the Microsoft Project version expands each element with durations, predecessors, resources, and responsibility assignments to support schedule development and critical path analysis. This enhanced level of detail ensures that the WBS functions not only as a scope definition tool but also as a fully integrated project management artifact that aligns the written structure with its operational implementation.

ID	WBS	Task Name	Responsibility	Duration	Predecessor	Resources	Description
1	1	Pre-Development Phase		340 days			
2	1.1	Community Engagement		60 days			
3	1.1.1	Conduct neighborhood listening sessions	Community Engagement Lead	25 days		Meeting space, facilitators	Facilitate open meetings to gather resident input on needs, priorities, and concerns.
4	1.1.2	Conduct interest gauge / pre-recruitment drive	Outreach Coordinator	15 days	1.1.1	Flyers, volunteers	Assess community interest in participation and identify potential volunteers or program contributors.
5	1.1.3	Facilitate stakeholder interviews	Project Manager	15 days	1.1.1	Scheduling tools	Interview key community leaders, partner organizations, and local agencies to refine project goals.
6	1.1.4	Compile community needs assessment	Analyst	10 days	1.1.2, 1.1.3	Data software	Synthesize engagement findings into a formal report that informs design and program decisions.
7	1.2	Governance Formation		70 days			
8	1.2.1	Establish community steering committee	Governance Lead	20 days	1.1.4	Community reps	Form a representative body responsible for oversight and decision making.
9	1.2.2	Develop cooperative or land trust framework	Legal Consultant	35 days	1.2.1	Legal templates	Define the ownership and governance structure for long term community control.
10	1.2.3	Approve governance charter	Steering Committee	15 days	1.2.2	Meeting space	Finalize and adopt the charter outlining roles, responsibilities, and operational guidelines.
11	1.3	Funding and Grant Preparation		70 days			
12	1.3.1	Prepare HUD CDBG application	Grant Writer	60 days	1.1.4	HUD guidelines	Develop and submit the Community Development Block Grant proposal.
13	1.3.2	Prepare CCED Sales Tax grant application	Grant Writer	50 days	1.1.4	CCED guidelines	Prepare the application for the Central City Economic Development Sales Tax program.
14	1.3.3	Prepare EDA Public Works grant application	Grant Writer	70 days	1.1.4	EDA guidelines	Complete the Economic Development Administration grant submission for capital funding.
15	1.4	Environmental & Architectural Preparation		210 days			
16	1.4.1	Conduct environmental review and testing	Environmental Consultant	45 days		Testing equipment	Perform assessments required for compliance with environmental regulations.
17	1.4.2	Conduct structural analysis and landscape analysis	Architect + Engineer	45 days	1.4.1	Site access	Evaluate building integrity and site conditions to inform design decisions.
18	1.4.3	Complete architectural and landscape schematic design	Architect	60 days	1.4.2	CAD software	Develop preliminary design concepts for interior, exterior, and site improvements.
19	1.4.4	Finalize construction documents	Architect	60 days	1.4.3	Drafting team	Produce detailed drawings and specifications required for permitting and construction.
20	2	Building Renovation Phase		400 days			
21	2.1	Site Preparation & Abatement		90 days			
22	2.1.1	Perform hazardous materials abatement	Abatement Contractor	40 days	1.4.4	PPE, disposal	Remove asbestos, lead, and other hazardous materials in compliance with safety standards.
23	2.1.2	Complete selective demolition	Demo Crew	30 days	2.1.1	Tools, dumpsters	Remove deteriorated or obsolete interior and exterior elements.
24	2.1.3	Secure site and install temporary utilities	General Contractor	20 days	2.1.2	Fencing, generators	Set up fencing, lighting, and temporary power to support construction activities.
25	2.2	Exterior Restoration		90 days			
26	2.2.1	Repair masonry and façade	Masonry Contractor	40 days	2.1.3	Scaffolding	Restore brickwork, stone, and exterior surfaces to preserve structural integrity.
27	2.2.2	Replace or restore windows and doors	Carpentry Contractor	30 days	2.2.1	Windows, tools	Install energy efficient or historically appropriate openings.
28	2.2.3	Repair roofing and drainage systems	Roofing Contractor	20 days	2.1.3	Roofing materials	Ensure the building envelope is weather tight and properly drained.
29	2.3	Mechanical, Electrical, and Plumbing (MEP)		60 days			
30	2.3.1	Install HVAC systems	HVAC Contractor	40 days	2.1.3	HVAC units	Install modern heating, ventilation, and air conditioning systems.
31	2.3.2	Upgrade electrical service and wiring	Electrician	60 days	2.1.3	Panels, wiring	Replace outdated wiring and increase electrical capacity.
32	2.3.3	Replace plumbing and install fixtures	Plumber	40 days	2.1.3	Pipes, fixtures	Install new piping, restrooms, and water service components.
33	2.4	Interior Construction & Finishes		120 days			
34	2.4.1	Build interior walls and program spaces	GC + Carpenters	60 days	2.3.1–2.3.3	Lumber, drywall	Construct classrooms, offices, labs, and multipurpose areas.
35	2.4.2	Install flooring, ceilings, and finishes	GC	40 days	2.4.1	Flooring, paint	Complete interior finishes consistent with program needs.
36	2.4.3	Install technology infrastructure and equipment	IT Contractor	20 days	2.4.1	Cabling, servers	Set up office, networking, computer labs, and audiovisual systems.
37	2.5	Final Inspections & Occupancy		40 days			
38	2.5.1	Conduct code compliance inspections	City Inspectors	15 days	2.4.2	Inspection tools	Coordinate inspections with local authorities to verify compliance.
39	2.5.2	Complete punch list corrections	GC	15 days	2.5.1	Tools, materials	Address remaining deficiencies identified during walkthroughs.
40	2.5.3	Obtain certificate of occupancy	City	10 days	2.5.2	Documentation	Secure official approval for building use.
41	3	Community Garden Development		65 days			
42	3.1	Site Preparation		30 days			
43	3.1.1	Clear and grade elevated field	Landscaping Crew	15 days		Tools	Remove debris and level the site for garden installation.
44	3.1.2	Install/repair fencing and access paths	Volunteers + Contractor	15 days	3.1.1	Fencing	Ensure safe and accessible entry to the garden area.
45	3.2	Garden Construction		20 days			
46	3.2.1	Build raised garden beds	Volunteers	10 days	3.1.2	Lumber	Construct durable beds for community planting.
47	3.2.2	Install irrigation system	Plumber + Volunteers	10 days	3.2.1	Irrigation kit	Set up water lines and drip irrigation for sustainable maintenance.
48	3.2.3	Deliver soil and compost	Supplier	5 days	3.2.1	Soil	Provide nutrient rich materials for planting.
49	3.3	Community Planting		15 days			
50	3.3.1	Organize volunteer planting days	Garden Coordinator	5 days	3.2.3	Seeds	Coordinate community participation in initial planting.
51	3.3.2	Plant initial crops	Volunteers	5 days	3.3.1	Plants	Begin cultivation of vegetables, herbs, and flowers.
52	3.3.3	Set up maintenance schedule	Coordinator	5 days	3.3.2	Calendar	Establish a volunteer rotation for ongoing care.
53	4	Administrative & Program Development		260 days			
54	4.1	Staffing & Volunteer Recruitment		90 days			
55	4.1.1	Recruit educators and program staff	HR Lead	60 days		Job postings	Identify and hire qualified personnel for educational and support programs.
56	4.1.2	Recruit volunteers for pantry and garden	Volunteer Coordinator	40 days		Outreach	Engage community members to support operations.
57	4.1.3	Conduct background checks and onboarding	HR	30 days	4.1.1, 4.1.2	Screening tools	Complete required screenings and orientation.
58	4.2	Program & Partnership Development		100 days			
59	4.2.1	Finalize partnership with Parks & Recreation	Director	40 days		Meetings	Confirm collaboration for youth and family programming.
60	4.2.2	Finalize partnership with Kansas City Public Library	Director	40 days		Meetings	Establish joint programming for literacy and community education.
61	4.2.3	Develop curricula for job training and community education	Program Lead	60 days	4.2.1, 4.2.2	Materials	Create structured learning materials for program delivery.
62	4.2.4	Coordinate donation drives	Volunteer Coordinator	30 days		Supplies	Organize community contributions for pantry and clothing closet operations.
63	4.3	Operational Readiness		70 days			
64	4.3.1	Develop policies and procedures	Director	50 days		Templates	Create operational guidelines for all program areas.
65	4.3.2	Create safety and emergency plans	Safety Officer	30 days		Manuals	Prepare protocols for emergency response and risk mitigation.
66	4.3.3	Prepare opening-day operations plan	Director	20 days	4.3.1, 4.3.2	Staff	Finalize staffing, logistics, and program schedules for launch.
67	5	Project Closeout		25 days			
68	5.1	Grand Opening		25 days			
69	5.1.1	Coordinate opening event	Director	20 days	2.5.3, 4.3.3	Event supplies	Plan and execute the public launch of the facility.
70	5.1.2	Launch job center and programming	Staff	5 days	5.1.1	Equipment	Begin delivery of workforce and educational services.
71	5.1.3	Launch pantry and closet	Volunteers	5 days	5.1.1	Supplies	Open food and clothing distribution services to the community.
72	5.1.4	Celebrate first garden harvest	Garden Coordinator	5 days	3.3.3	Produce	Host a community event marking the first successful growing cycle.

Figure 1

Work Breakdown Structure for the Sanford B. Ladd School Redevelopment Project

Conclusion

The completed WBS provides a structured representation of the major deliverables required to transform the Sanford B. Ladd School into a functional community resource hub. By breaking the project into detailed activities across pre-development, renovation, garden construction, and administrative preparation, the WBS strengthens scope clarity and supports the coordination of work that must comply with federal funding guidelines and local redevelopment requirements (City of Kansas City, Missouri, 2024; U.S. Economic Development Administration, 2023). This level of decomposition also aligns with the technical expectations outlined in the project's architectural and environmental documentation, ensuring that the redevelopment effort remains grounded in realistic construction and design parameters (Drexel Tech, 2026; LocalCode, n.d.). In addition, the expanded WBS developed in Microsoft Project incorporates durations, dependencies, and resource assignments, allowing the project team to validate sequencing, identify the critical path, and confirm that the 36 to 38 month timeline established in the proposal is achievable. As the project progresses, the WBS will serve as a reference point for monitoring performance, managing risks, and ensuring that all components of the redevelopment effort remain aligned with community needs and long-term goals.

References

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WBS Dictionary

Project: Redevelopment of the Sanford B. Ladd School into a Community Resource Hub

1.0 Pre Development Phase

1.1 Community Engagement

1.1.1 Conduct neighborhood listening sessions

Facilitate open meetings to gather resident input on needs, priorities, and concerns.

1.1.2 Conduct interest gauge / pre-recruitment drive

Assess community interest in participation and identify potential volunteers or program contributors.

1.1.3 Facilitate stakeholder interviews

Interview key community leaders, partner organizations, and local agencies to refine project goals.

1.1.4 Compile community needs assessment

Synthesize engagement findings into a formal report that informs design and program decisions.

1.2 Governance Formation

1.2.1 Establish community steering committee

Form a representative body responsible for oversight and decision making.

1.2.2 Develop cooperative or land trust framework

Define the ownership and governance structure for long-term community control.

1.2.3 Approve governance charter

Finalize and adopt the charter outlining roles, responsibilities, and operational guidelines.

1.3 Funding and Grant Preparation

1.3.1 Prepare HUD CDBG application

Develop and submit the Community Development Block Grant proposal.

1.3.2 Prepare CCED Sales Tax grant application

Prepare the application for the Central City Economic Development Sales Tax program.

1.3.3 Prepare EDA Public Works grant application

Complete the Economic Development Administration grant submission for capital funding.

1.4 Environmental and Architectural Preparation

1.4.1 Conduct environmental review and testing

Perform assessments required for compliance with environmental regulations.

1.4.2 Conduct structural and landscape analysis

Evaluate building integrity and site conditions to inform design decisions.

1.4.3 Complete architectural and landscape schematic design

Develop preliminary design concepts for interior, exterior, and site improvements.

1.4.4 Finalize construction documents

Produce detailed drawings and specifications required for permitting and construction.

2.0 Building Renovation Phase

2.1 Site Preparation and Abatement

2.1.1 Perform hazardous materials abatement

Remove asbestos, lead, and other hazardous materials in compliance with safety standards.

2.1.2 Complete selective demolition

Remove deteriorated or obsolete interior and exterior elements.

2.1.3 Secure site and install temporary utilities

Set up fencing, lighting, and temporary power to support construction activities.

2.2 Exterior Restoration

2.2.1 Repair masonry and façade

Restore brickwork, stone, and exterior surfaces to preserve structural integrity.

2.2.2 Replace or restore windows and doors

Install energy efficient or historically appropriate openings.

2.2.3 Repair roofing and drainage systems

Ensure the building envelope is weather tight and properly drained.

2.3 Mechanical, Electrical, and Plumbing (MEP)

2.3.1 Install HVAC systems

Install modern heating, ventilation, and air conditioning systems.

2.3.2 Upgrade electrical service and wiring

Replace outdated wiring and increase electrical capacity.

2.3.3 Replace plumbing and install fixtures

Install new piping, restrooms, and water service components.

2.4 Interior Construction and Finishes

2.4.1 Build interior walls and program spaces

Construct classrooms, offices, labs, and multipurpose areas.

2.4.2 Install flooring, ceilings, and finishes

Complete interior finishes consistent with program needs.

2.4.3 Install technology infrastructure and equipment

Set up office, networking, computer labs, and audiovisual systems.

2.5 Final Inspections and Occupancy

2.5.1 Conduct code compliance inspections

Coordinate inspections with local authorities to verify compliance.

2.5.2 Complete punch list corrections

Address remaining deficiencies identified during walkthroughs.

2.5.3 Obtain certificate of occupancy

Secure official approval for building use.

3.0 Community Garden Development

3.1 Site Preparation

3.1.1 Clear and grade elevated field

Remove debris and level the site for garden installation.

3.1.2 Install or repair fencing and access paths

Ensure safe and accessible entry to the garden area.

3.2 Garden Construction

3.2.1 Build raised garden beds

Construct durable beds for community planting.

3.2.2 Install irrigation system

Set up water lines and drip irrigation for sustainable maintenance.

3.2.3 Deliver soil and compost

Provide nutrient rich materials for planting.

3.3 Community Planting

3.3.1 Organize volunteer planting days

Coordinate community participation in initial planting.

3.3.2 Plant initial crops

Begin cultivation of vegetables, herbs, and flowers.

3.3.3 Set up maintenance schedule

Establish a volunteer rotation for ongoing care.

4.0 Administrative and Program Development

4.1 Staffing and Volunteer Recruitment

4.1.1 Recruit educators and program staff

Identify and hire qualified personnel for educational and support programs.

4.1.2 Recruit volunteers for pantry and garden

Engage community members to support operations.

4.1.3 Conduct background checks and onboarding

Complete required screenings and orientation.

4.2 Program and Partnership Development

4.2.1 Finalize partnership with Parks and Recreation

Confirm collaboration for youth and family programming.

4.2.2 Finalize partnership with Kansas City Public Library

Establish joint programming for literacy and community education.

4.2.3 Develop curricula for job training and community education

Create structured learning materials for program delivery.

4.2.4 Coordinate donation drives

Organize community contributions for pantry and clothing closet operations.

4.3 Operational Readiness

4.3.1 Develop policies and procedures

Create operational guidelines for all program areas.

4.3.2 Create safety and emergency plans

Prepare protocols for emergency response and risk mitigation.

4.3.3 Prepare opening day operations plan

Finalize staffing, logistics, and program schedules for launch.

5.0 Project Closeout

5.1 Grand Opening

5.1.1 Coordinate opening event

Plan and execute the public launch of the facility.

5.1.2 Launch job center and programming

Begin delivery of workforce and educational services.

5.1.3 Launch pantry and closet

Open food and clothing distribution services to the community.

5.1.4 Celebrate first garden harvest

Host a community event marking the first successful growing cycle.